

The Illusion of Competence

Discussion of Anthropic Research: “How AI assistance impacts the formation of coding skills”

AI path

ask AI → get code → task passes

VS.

Human path

predict → try → fail → debug → explain → retain

Which path did you use the last time AI helped you code?

Anthropic Research article + Shen & Tamkin, 2026

The visible win hides the invisible cost

WHAT ORGANIZATIONS SEE

faster drafts
completed tickets
fewer syntax errors



WHAT MAY BE MISSING

weaker mental model
less debugging practice
less rejection of plausible
wrong code

“How many of us have approved or shipped code this week that we could not fully explain line by line?”

How using AI to program impact learning

It asks what happens when AI is used while people are learning.

PRIOR PRODUCTIVITY STORY

AI can speed up familiar work;
Anthropic cites prior evidence of
large task-level productivity gains.

THIS STUDY

developers learn an unfamiliar async
library, Trio, with or without an AI
sidebar.

CORE QUESTION

Does AI provide a shortcut to both
efficiency and skill development?

The article is about skill formation under AI assistance, not AI productivity in general.

The Experiment

The design separates “finished the task” from “retained the skill.”

1 Recruit

52 Python developers;
Trio unfamiliar

2 Calibrate

warm-up task without AI
(10mins)

3 Learn

two Trio tasks; AI sidebar
or no AI (35 mins)

4 Evaluate

final quiz with **no AI** (25
mins)

AI CONDITION

chat-based coding assistant in the sidebar; it could see current code and generate code

NO-AI CONDITION

same learning materials and coding tasks, but no assistant during the Trio stage

What exactly did they ask developers to learn?

Trio was a deliberately unfamiliar async library, not a Python syntax test.

Task 1

Write a timer that prints each second while other functions run.

TARGET CONCEPTS

nurseries · starting tasks · concurrent functions

Task 2

Implement record retrieval that handles missing-record errors.

TARGET CONCEPTS

error handling · memory channels · result flow

The task simulates workplace onboarding: a short tutorial, starter code, then a small feature using a new library.

The Evaluation Quiz

Anthropic frames mastery as supervising code after the assistant is gone.

14

questions

27 total points

4

assessment types

from CS education

0

AI access

during the quiz

DEBUGGING

identify and diagnose errors

CODE READING

understand and verify code

CODE WRITING

select or write the correct approach to write code

CONCEPTUAL

core principles behind tools and library

How did they conclude there was a learning penalty?

Random assignment makes the comparison: same task, different access to AI.

GROUPS

- 26 AI participants
- 26 no-AI participants

PRODUCTIVITY OUTCOME

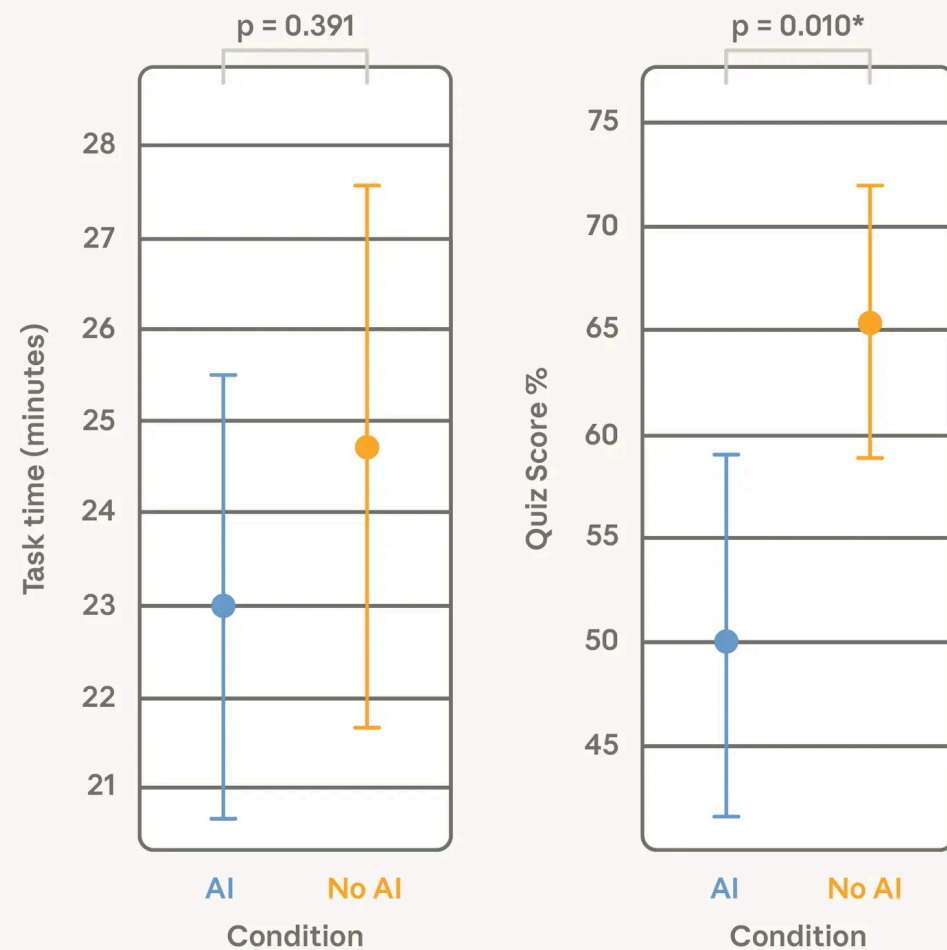
completion time difference: about two minutes
not significant (p=0.391)

LEARNING OUTCOME

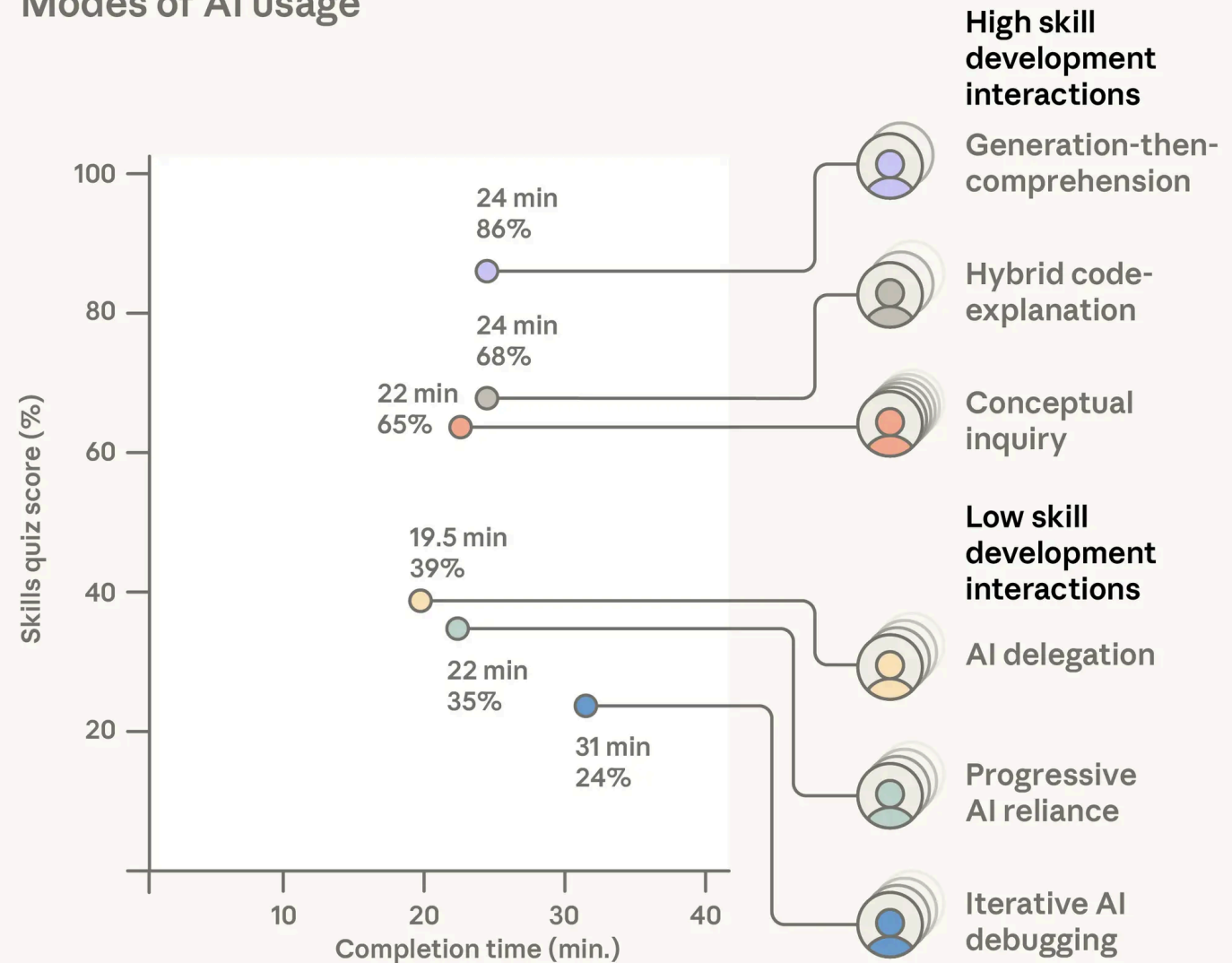
quiz gap: 4.15 points on a 27-point quiz
significant (d=0.738, p=0.010)

How AI assistance impacts coding speed and skill formation

AI assistance: treatment effect on coding speed and knowledge score

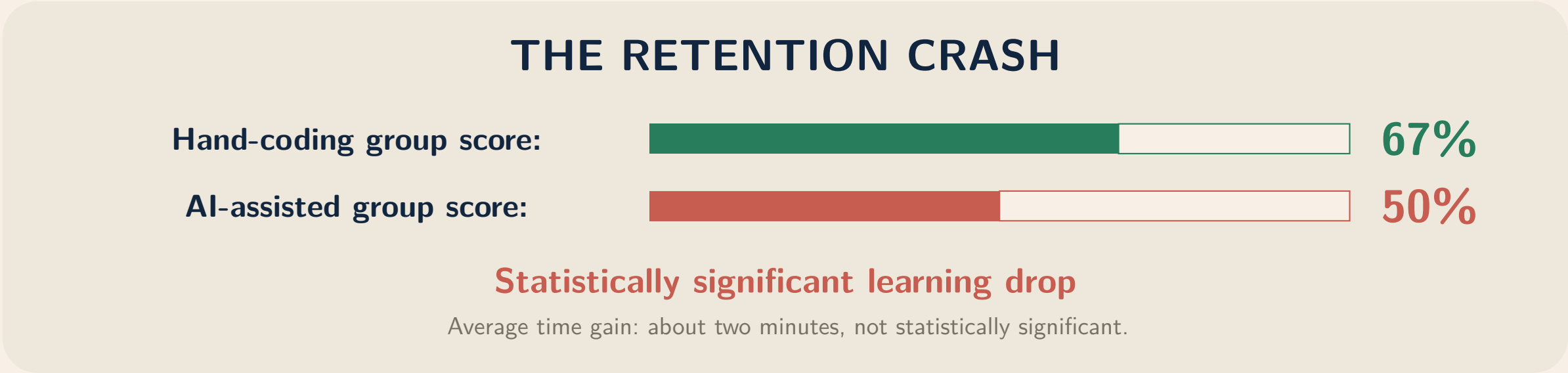


Modes of AI usage



The Retention Crash

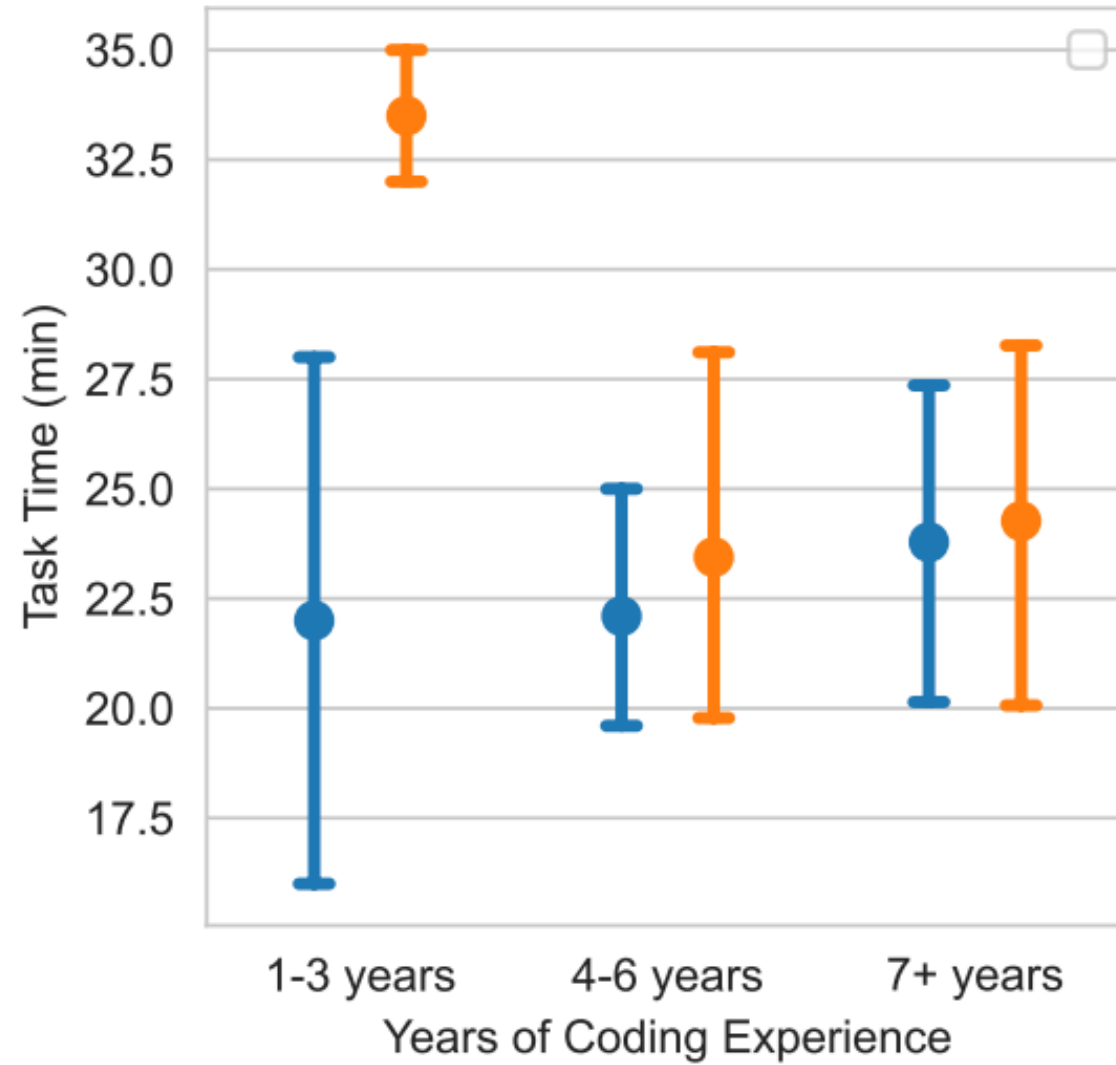
Lower scores.



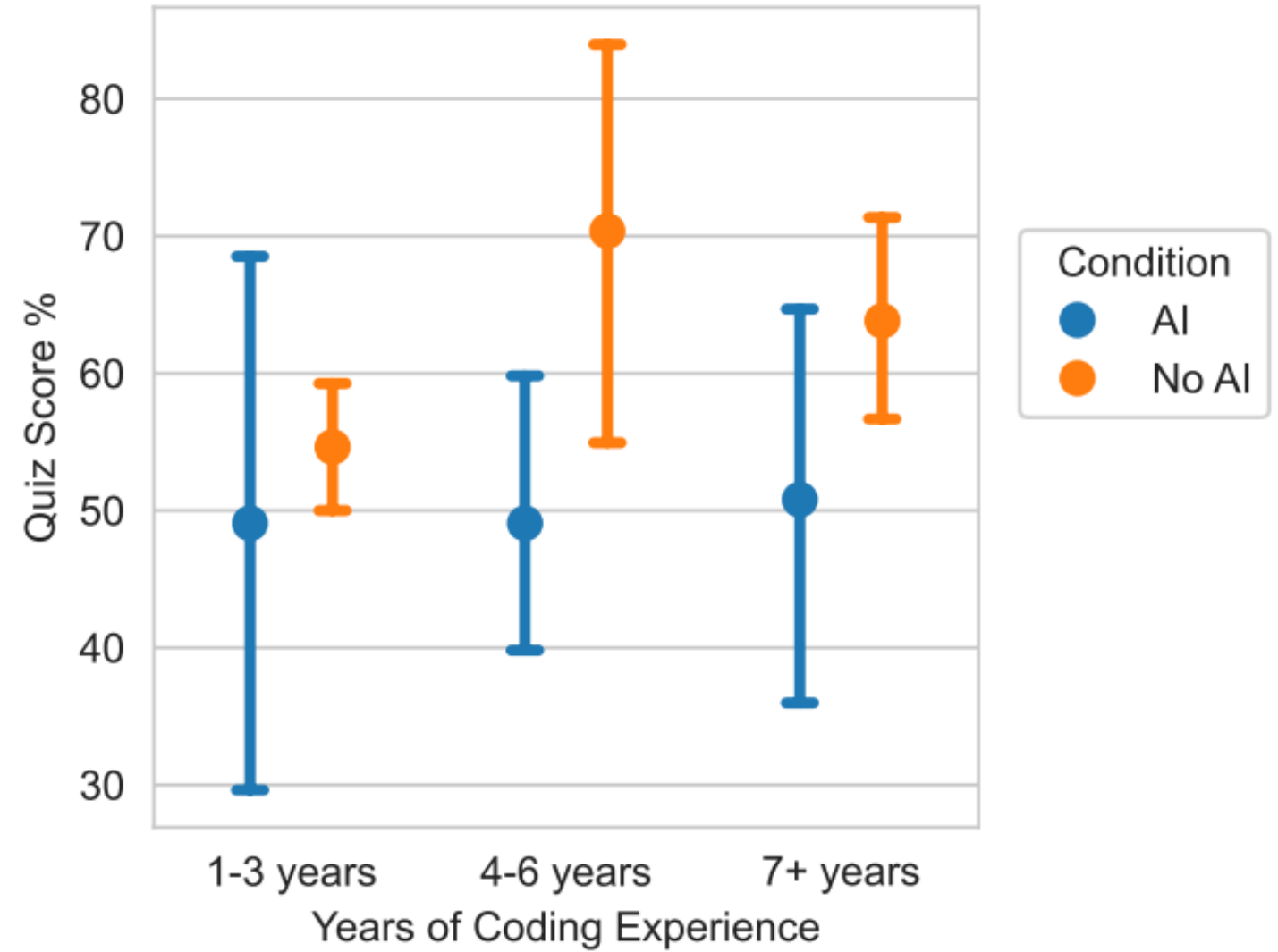
A finished task did not translate into retained mastery.

Main Study: Skill Development by Years of Coding Experience

Task Time by Years of Coding Experience



Quiz Score by Years of Coding Experience



Debugging is where the gap bites

No-AI learners hit errors



they repaired those errors



repair trained judgment

OBSERVED PATTERN

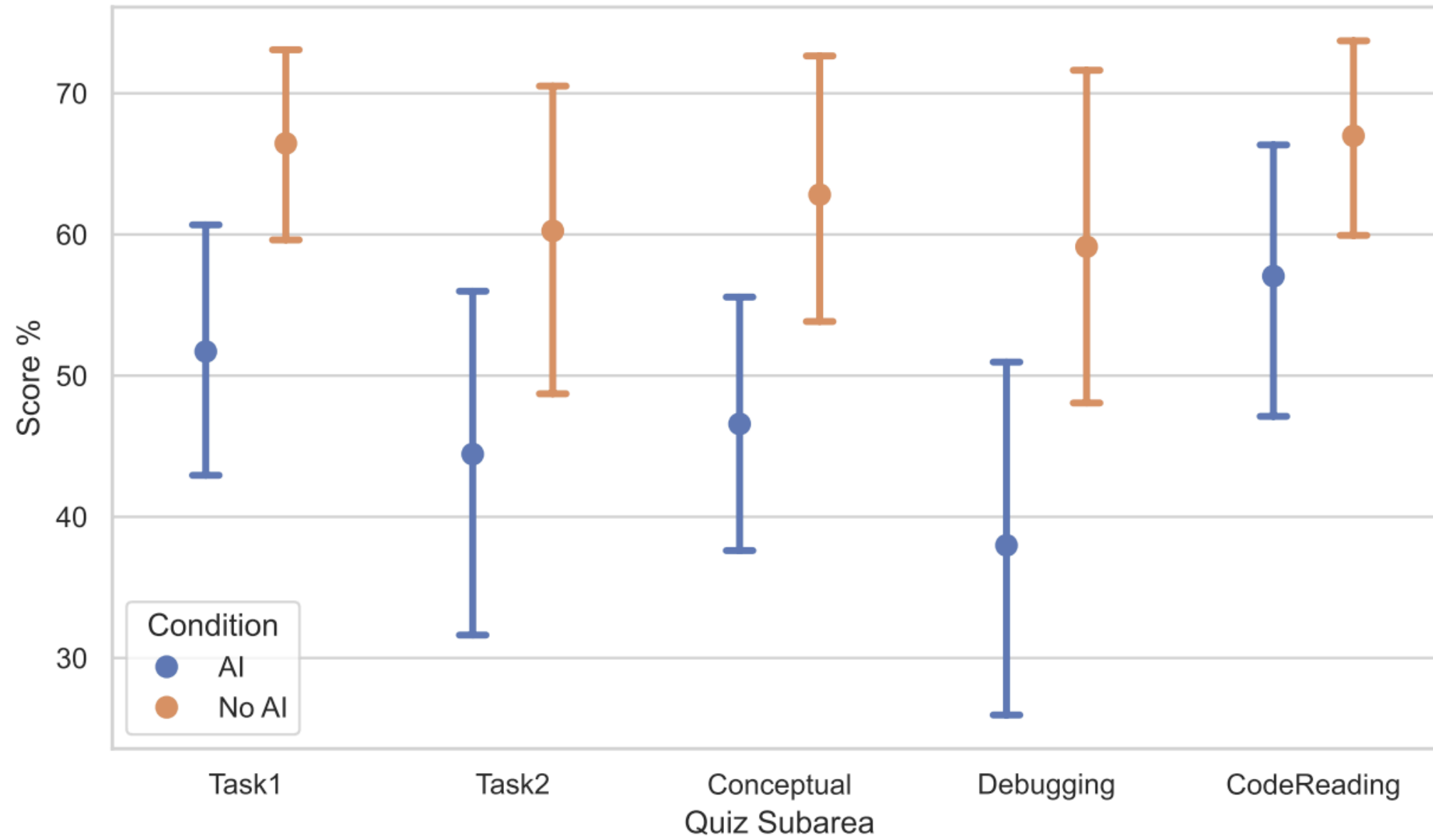
Largest gap: debugging.

Smallest gap: code reading.

CAVEAT

Plausible mechanism, not proven mediation.

Quiz Score % by Subarea and Condition



The article's six AI interaction modes

The result is not “AI bad”; it is “interaction pattern matters.”

LOW-SCORING PATTERNS

AI delegation n=4: asks AI to write the task

Progressive reliance n=4: do task 1, then ask AI for task 2

Iterative AI debugging n=4: asks AI to solve or verify failures

Average quiz scores below 40%; more cognitive offloading.

HIGH-SCORING PATTERNS

Generation-then-comprehension n=2: asks follow-up questions

Hybrid code-explanation n=3: asks for code plus explanations

Conceptual inquiry n=7: asks concepts, codes independently

Average quiz scores at or above 65%; comprehension stays active.

The article's qualitative analysis is descriptive; it does not prove that the interaction modes causally determine scores.

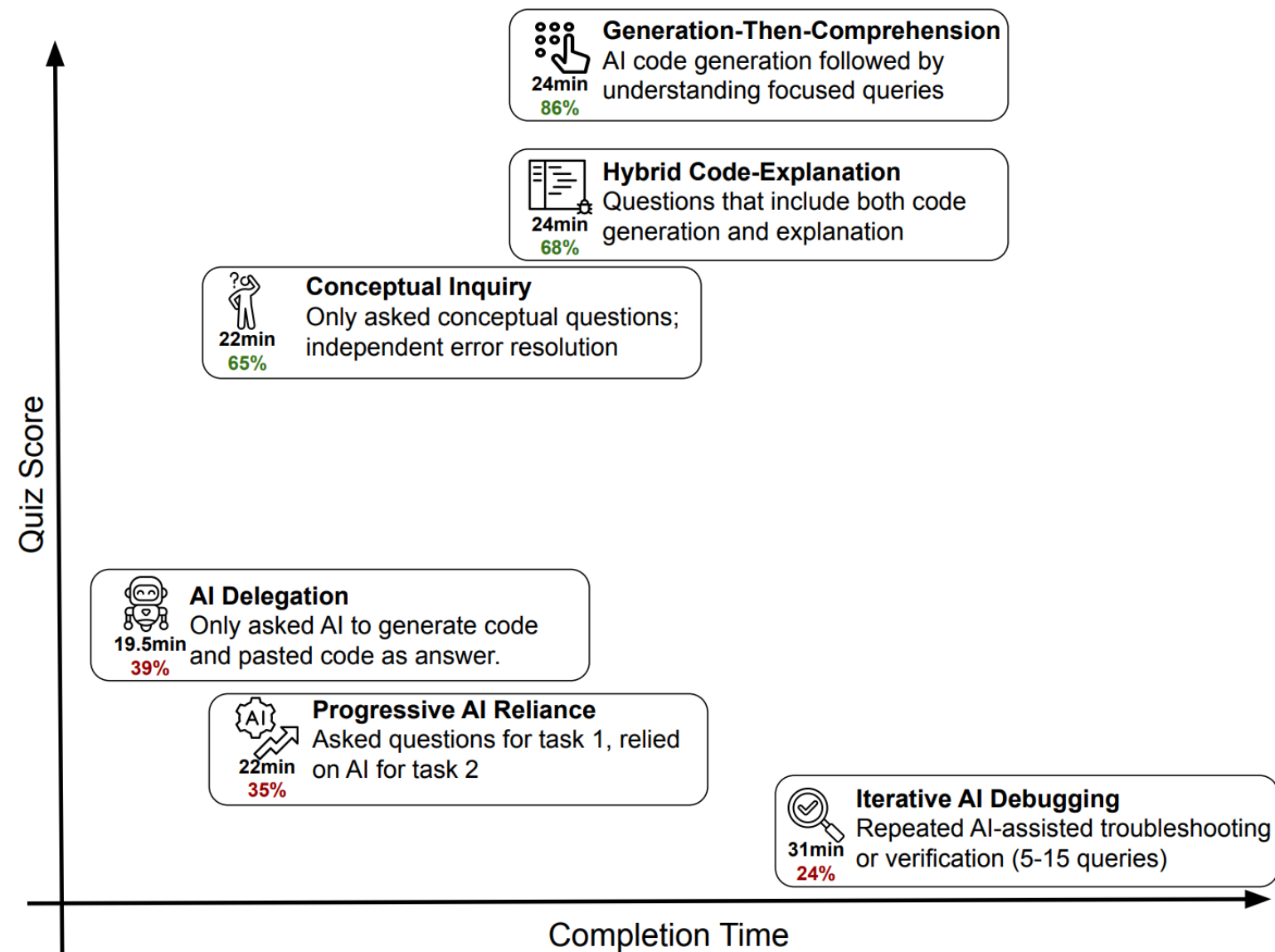


Figure 11: The 6 AI interaction personas in the treatment (AI) condition from our study with average completion times and quiz scores.

Same assistant. Different practice loop.

artifact path

DELEGATION

1. User: write the solution
2. AI: here is code
3. User: paste / pass

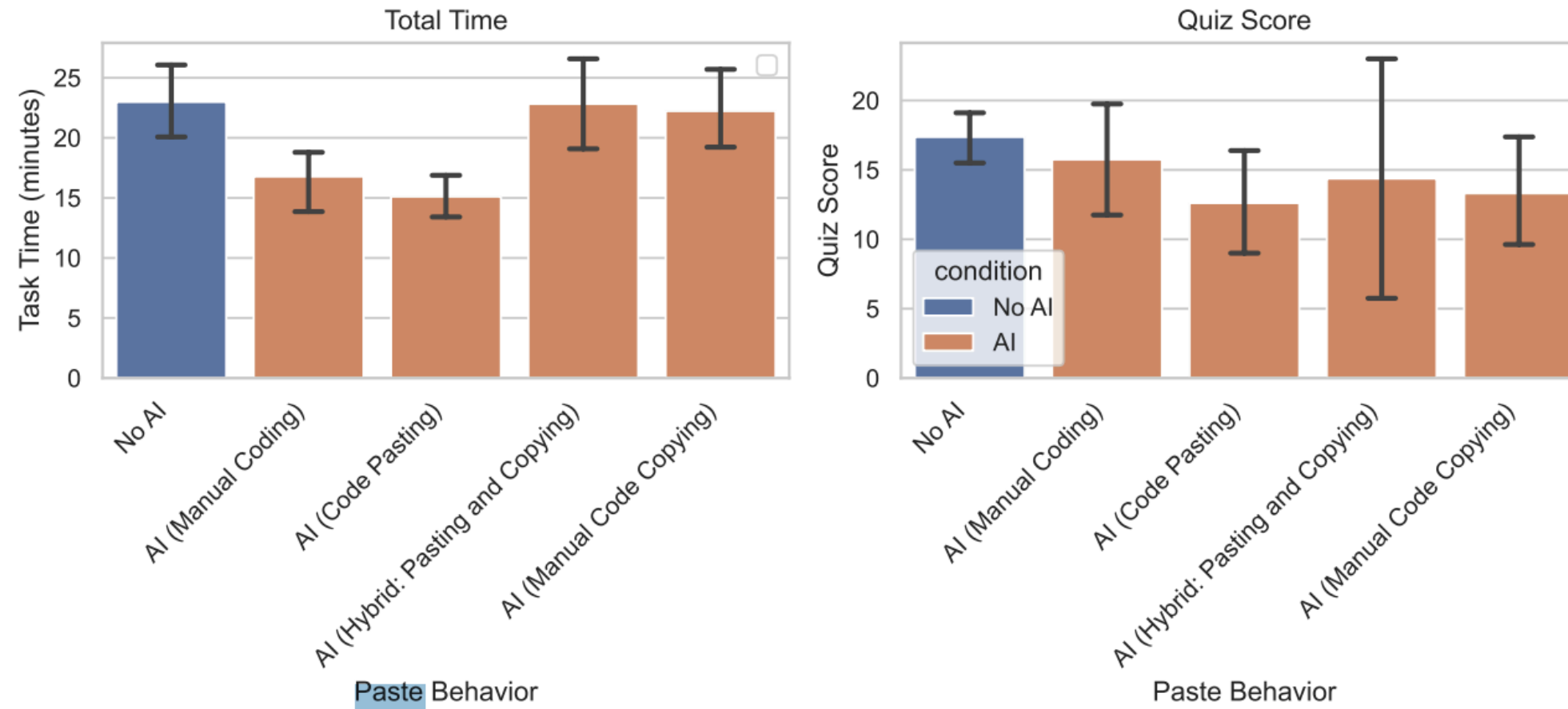
vs.

learning path

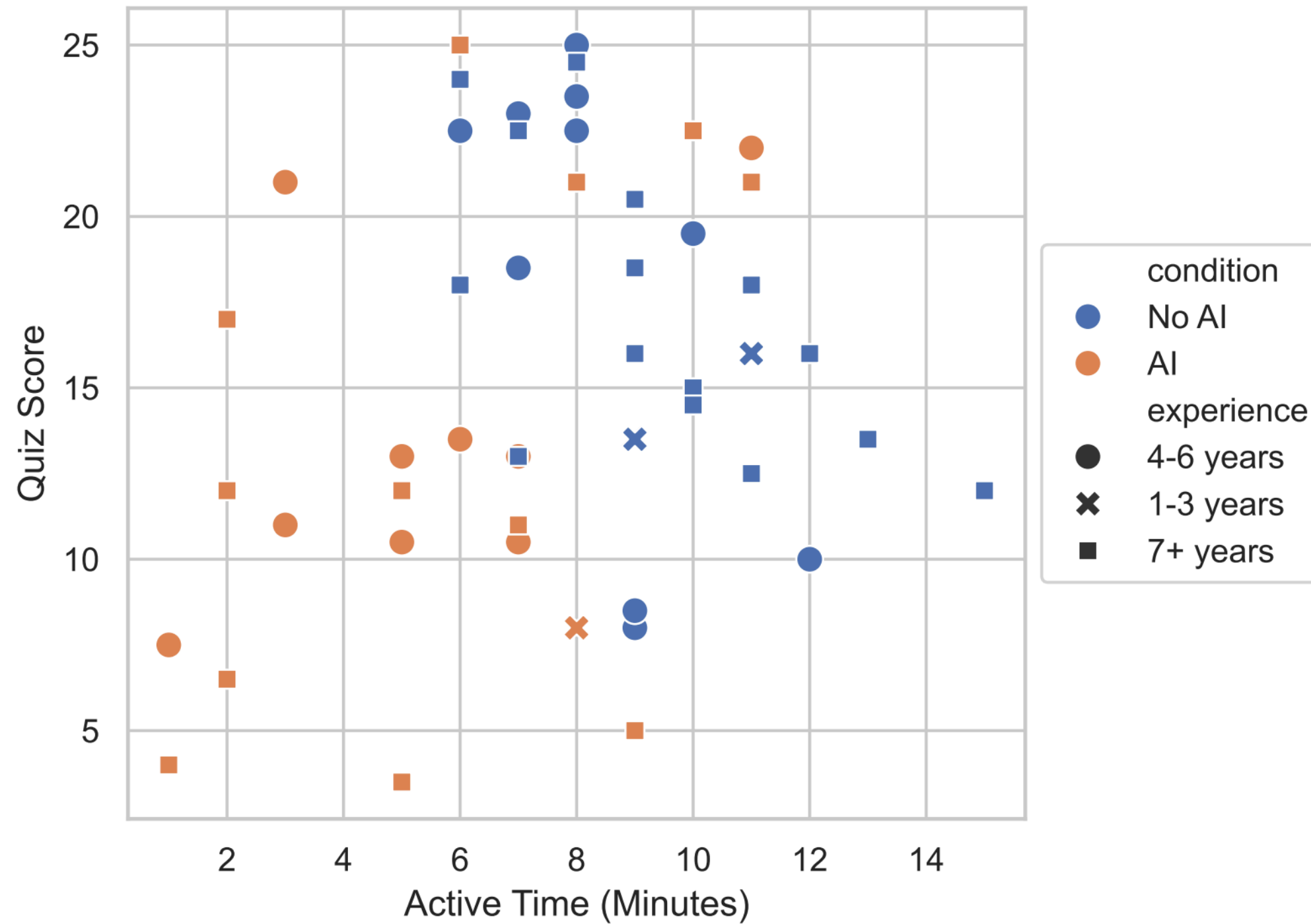
LEARNING

1. User: asks concept
2. User: predicts behavior
3. User: tries, hits error

Paste Behavior vs Task Time and Quiz Score



Coding Activity: Active Time vs Quiz Score



The persona labels are not controlled comparisons

They describe what people did; they do not isolate why.

WHAT THE LABELS SHOW

AI delegation scored low.

Conceptual inquiry scored high.

Useful pattern: cognitive engagement seems important.

WHAT THEY DO NOT CONTROL

Years of experience.

Prior async knowledge.

Prompting skill.

Whether the learner was already stuck.

Paper grounding: the six interaction modes are qualitative clusters from small groups (n=2–7), not randomized conditions with variables independently controlled.

A little personal experience...

Delegation is not helping me to learn...

ASK

AI to do a general code path profiling of a gRPC system: tonic.



INSPECT

Comparisons were confounded; hot-path timers changed the system.



FAILURE

Wrong regimes; instrumentation-off throughput; bad advice on how to understand the performance gap

Directed debugging kept me in the learning loop

Struggling is where I learn...

MY DIRECTION

Focus on DSA alignment inside
coroutine-frame storage:
64-byte descriptor, 32-byte
completion.

CLAUDE'S WORK

inspect layout
challenge `alignas()`
trace where addresses are formed

FIX

over-allocate
compute aligned pointers once
cache hot-path addresses

DEBUGGING MODEL

`(base + align - 1) & ~(align - 1)` preserved the
hardware contract when coroutine storage made naive
alignment suspect.

This is closer to conceptual inquiry than delegation:
Claude searched; I supplied the model to test.

One question to end on

If AI can produce the artifact, what should undergraduate CS education do to help students learn?

DISCUSSION

What should we still make students struggle through—and why?

AI-enhanced productivity is not a shortcut to competence

If future oversight depends on human expertise, our tools and habits must preserve the practice loop.

PRIMARY

Shen, Judy Hanwen, and Alex Tamkin. “How AI Impacts Skill Formation.” arXiv:2601.20245, 2026.

ARTICLE

Anthropic Research. “How AI assistance impacts the formation of coding skills.” 2026.

RELATED

Storey, Margaret-Anne. “How Generative and Agentic AI Shift Concern from Technical Debt to Cognitive Debt.” 2026.
margaretstorey.com/blog/2026/02/09/cognitive-debt/

RELATED

Sarkar, Advait. “Intention Is All You Need.” arXiv:2410.18851, 2024. arxiv.org/abs/2410.18851

VIDEO

Anthropic Research on AI and skill formation. YouTube reference: bp8dwCJF6iw.

CONTEXT

Prior AI-productivity, critical-thinking, and cognitive-offloading work cited in the Anthropic article.